

MULTI-TOUCH MARKETING ATTRIBUTION & ROI DASHBOARD

Executive Project Report

Submitted By

Yusuf Murtaza Bahrainwala

Project Domain

Marketing Analytics & Business Intelligence

Project Description

This project presents an end-to-end Marketing Analytics solution for evaluating the effectiveness of multi-channel digital marketing campaigns. By integrating advertising spend, customer touch-points, and conversion data, the project implements **First-Touch, Last-Touch, and Linear Attribution Models** to accurately distribute conversion credit across the customer journey.

The solution leverages **Python** for data preprocessing, **PostgreSQL** for advanced SQL-based attribution modeling, and **Power BI** for interactive dashboard development. It provides comprehensive insights into campaign performance, customer acquisition, marketing return on investment, and conversion funnel analysis, enabling data-driven marketing decisions.

Technology Stack

- Python (Pandas)
- SQL (PostgreSQL)
- Power Query & DAX
- Microsoft Power BI

Key Business Metrics

- Total Revenue
 - Total Marketing Spend
 - Total Conversions
 - Return on Ad Spend (ROAS)
 - Customer Acquisition Cost (CAC)
 - Click-Through Rate (CTR)
 - Cost Per Click (CPC)
 - Attributed Revenue
 - Customer Journey & Funnel Analysis
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Project Timeline

June 5, 2026 to July 11, 2026

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2. Executive Summary

Business Problem

Organizations invest significant budgets across multiple digital marketing channels, including Google Ads, Meta Ads, LinkedIn, TikTok, and Email Marketing, to acquire and retain customers. However, modern customer journeys involve multiple interactions before a conversion occurs, making it difficult to accurately identify which marketing channels contribute most to business outcomes. Traditional single-touch attribution models often assign all conversion credit to the first or last interaction, resulting in incomplete performance evaluation and inefficient marketing budget allocation.

Project Objective

The primary objective of this project was to develop a comprehensive Multi-Touch Marketing Attribution solution capable of integrating marketing spend, customer touchpoints, and conversion data into a unified analytical framework. The solution implements **First-Touch**, **Last-Touch**, and **Linear Attribution** models to fairly distribute conversion credit across the customer journey and provide actionable insights into campaign effectiveness, customer acquisition efficiency, and marketing return on investment.

Solution Overview

The project follows an end-to-end analytics workflow beginning with data preprocessing in Python, followed by SQL-based data transformation and attribution modeling in PostgreSQL. The transformed data was structured into a Star Schema consisting of fact and dimension tables to support efficient analytical querying. Finally, interactive dashboards were developed in Microsoft Power BI using DAX measures to visualize key marketing KPIs, campaign performance, attribution comparisons, and customer journey analysis.

Business Objectives

The solution was designed to achieve the following objectives:

- Accurately attribute conversion revenue across multiple marketing touchpoints.
 - Evaluate campaign performance using multiple attribution models.
 - Measure key marketing KPIs including Revenue, ROAS, CAC, CTR, and CPC.
 - Analyze customer journeys and identify conversion funnel drop-off points.
 - Enable data-driven marketing budget allocation through interactive dashboards.
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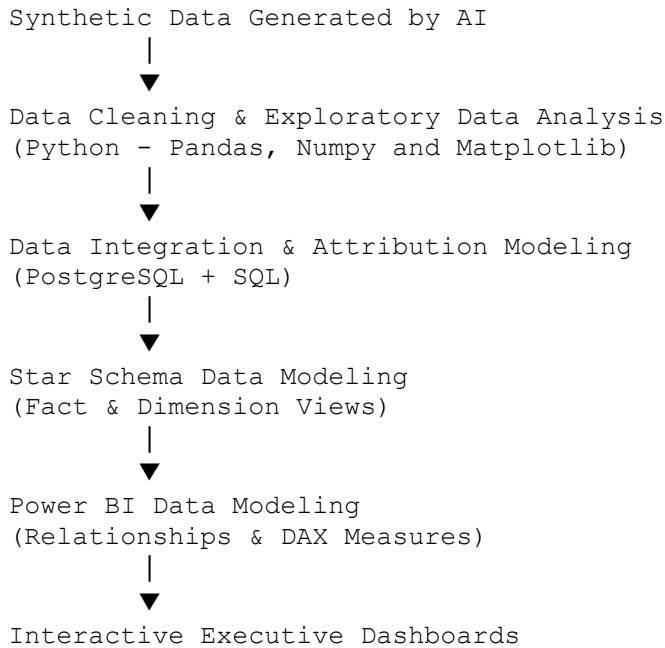
Expected Business Impact

The developed solution enables marketing teams to gain a comprehensive understanding of campaign performance across multiple digital channels. By comparing attribution models and analyzing customer journeys, decision-makers can optimize advertising budgets, improve campaign efficiency, reduce customer acquisition costs, and maximize marketing return on investment through evidence-based decision-making.

3. Project Methodology & Data Pipeline

Project Workflow

This project follows a structured end-to-end analytics workflow that transforms raw marketing data into interactive business intelligence dashboards. Each stage focuses on preparing, modeling, and analyzing data to generate actionable marketing insights.



Dataset Overview

The project integrates three independent datasets representing different stages of the marketing lifecycle.

1. Ad Spend Dataset

Key Fields - Campaign ID, Campaign Name, Platform, Channel, Spend, Clicks, Impressions.

Purpose - Used to calculate campaign investment, advertising efficiency, CPC, CTR, ROAS and CAC

2. User Touchpoints Dataset

Key Fields - User ID, Session ID, Timestamp, Source, Medium, Campaign, Event Type.

Purpose - Used to construct customer journey table and sequence touchpoints for attribution modelling.

3. Conversion Dataset

Key Fields - Conversion ID, User ID, Conversion Timestamp, Revenue.

Purpose - Provides the final business outcome required for attribution and revenue allocation.

Data Processing & Attribution Logic

The datasets were transformed and integrated using **PostgreSQL**. Advanced SQL techniques, including JOIN operations, Window Functions, and Views, were used to build a complete customer journey for each converted user.

Three attribution models were implemented:

- **First-Touch** – Assigns full conversion credit to the customer's first interaction.
- **Last-Touch** – Assigns full conversion credit to the final interaction before conversion.
- **Linear Attribution** – Evenly distributes conversion credit across all customer touchpoints.

These models enable comparison of different marketing attribution strategies and provide a more comprehensive understanding of campaign performance.

Data Modeling

To support efficient reporting and analysis, the transformed data was organized into a **Star Schema**.

Fact Table

Fact Attribution

- Attribution Revenue
- Attributed Conversions
- Attribution Model Type
- Campaign ID
- Campaign Name
- Date

Dimension Tables

Dim Campaign

- Campaign Id
- Campaign Name
- Platform
- Channel

Dim Date

- Date
- Weekday
- Month
- Quarter
- Year

Dim User Journey

- User ID
- Campaign
- Conversion Timestamp
- Event Type
- Medium
- Revenue
- Source
- Session ID

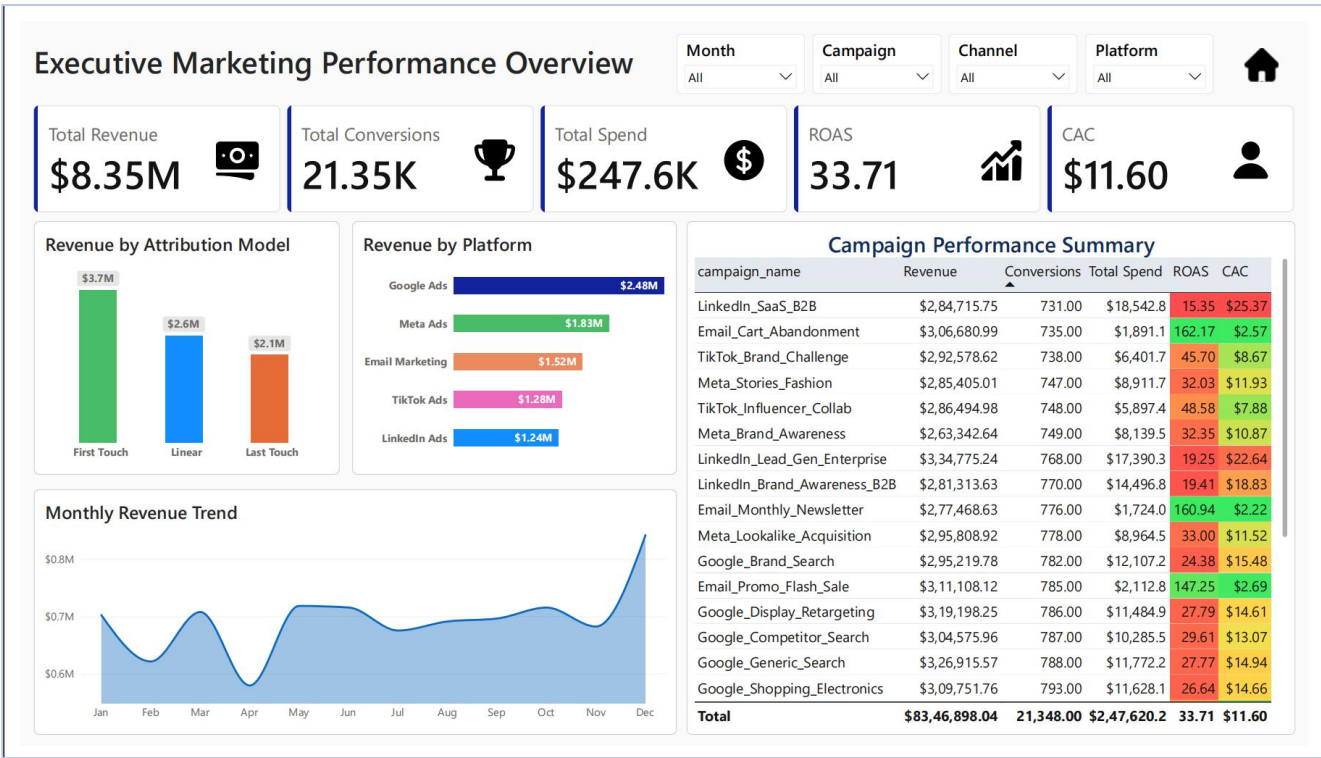
Also KPIs like ROAS, CAC, CTR, Attributed Revenue, Clicks, Impressions, CPC were calculated.

4. Dashboard Overview

Marketing Analytics Dashboards

To support data-driven marketing decisions, four interactive dashboards were developed in Microsoft Power BI. Each dashboard focuses on a specific aspect of marketing performance, allowing stakeholders to analyze campaign effectiveness, attribution strategies, customer journeys, and return on marketing investment.

Dashboard 1 – Executive Marketing Performance Overview



Purpose

This dashboard provides a consolidated overview of key marketing performance indicators, enabling executives to monitor overall business performance and evaluate marketing effectiveness at a glance.

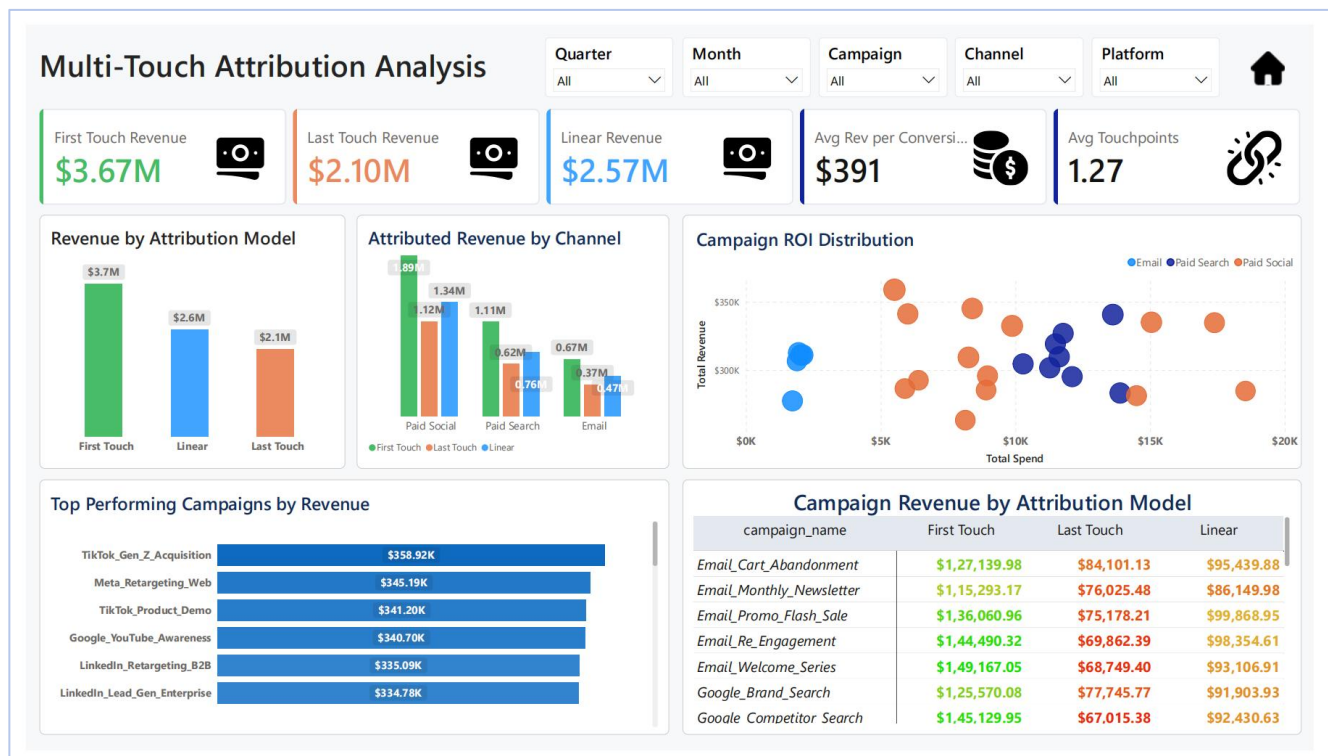
Key Highlights

- Displays overall Revenue, Marketing Spend, Conversions, ROAS, and CAC.
- Compares revenue and conversions performance across channels.
- Monitors monthly revenue trends.

Key Insights Derived

- Google Ads Platform generated the highest revenue followed by Meta Ads and Email Marketing, indicating Google is the primary acquisition channel.
- December records highest revenue generation while April records the lowest.
- Email marketing delivers highest return for every dollar invested with efficient Customer acquisition.

Dashboard 2 – Multi-Touch Attribution Analysis



Purpose

This dashboard compares different attribution models to demonstrate how conversion credit is distributed across customer touchpoints, helping marketing teams understand the influence of each channel throughout the customer journey.

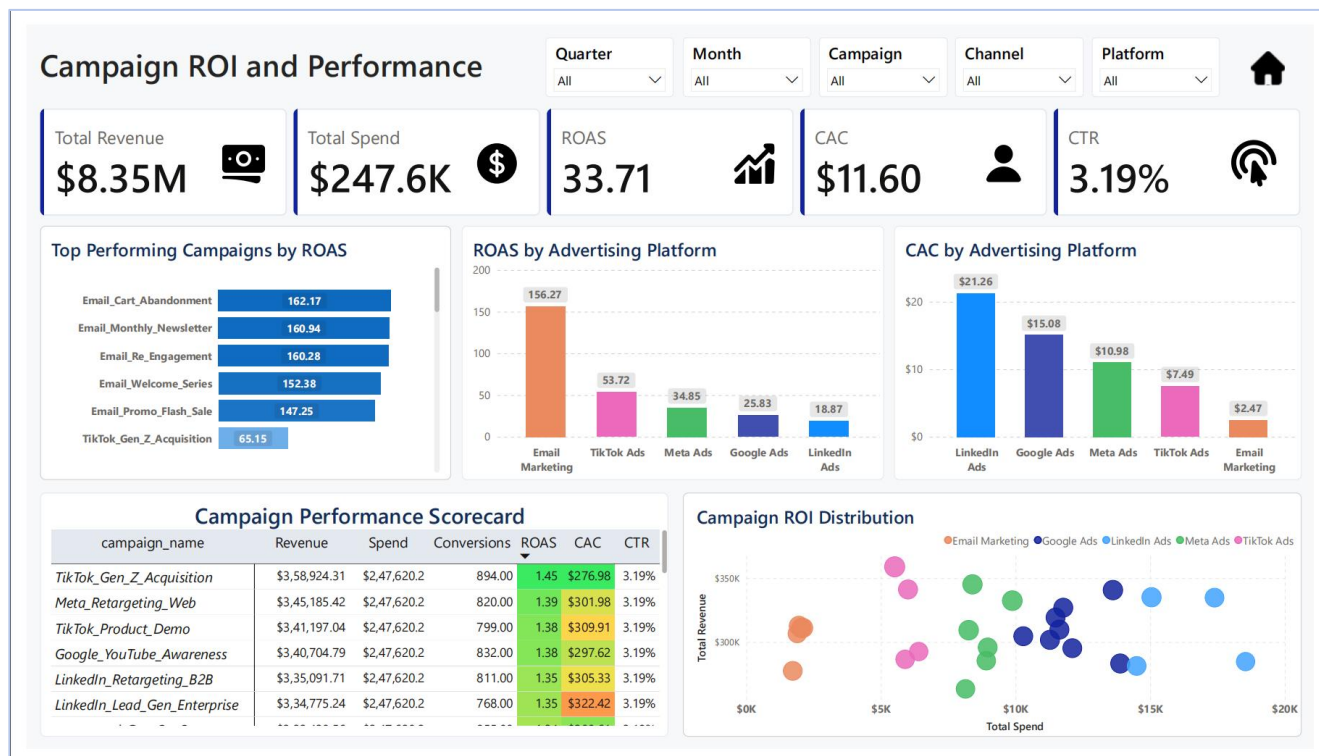
Key Highlights

- Compares First-touch, Last-touch and Linear Attribution Models.
- Evaluates attributed revenue across marketing channels.
- Identifies top performing campaigns under different attribution models.
- Summarizes campaign revenue for different attribution models.

Key Insights Derived

- The First-Touch model attributes the highest revenue (\$3.67M), indicating that the initial customer interaction plays a significant role in influencing conversions.
- Paid Social is the strongest revenue-generating channel across the customer journey, indicating it is effective in both attracting customers and assisting conversions.
- A small group of campaigns from different platforms contributes a significant share of total attributed revenue, demonstrating that campaign performance is concentrated rather than evenly distributed.
- Some LinkedIn Campaigns with higher spend generate comparably less revenue whereas some from TikTok and Google generate high revenue with moderate spend, indicating higher advertising spend does not necessarily guarantee higher revenue.

Dashboard 3 – Campaign ROI and Performance Analysis



Purpose

This dashboard focuses on campaign-level performance by analyzing profitability, advertising efficiency, and return on marketing investment through ROAS and CAC calculations.

Key Highlights

- Displays overall Revenue, Marketing Spend, ROAS, CAC and CTR.
- Compares campaign spend against attributed revenue.
- Highlights high performing campaigns by ROAS.
- Analyzes Platform distribution by ROAS and CAC.
- Summarizes campaign performance matrix for decision-making.

Key Insights Derived

- Email Marketing significantly outperforms all other platforms by generating the highest return on ad spend while acquiring customers at the lowest cost. This indicates that email campaigns are the most cost-effective marketing channel.
- LinkedIn campaigns shows the highest acquisition cost while generating the lowest return, suggesting lower marketing efficiency compared to other platforms.
- Email Marketing campaigns consistently generate highest returns, indicating a highly effective campaign strategy.
- The marketing portfolio generates approximately \$33.71 in revenue for every \$1 spent on advertising, indicating strong overall marketing performance.

Dashboard 4 – Customer Journey and Funnel Analysis



Purpose

This dashboard analyzes customer interactions across the conversion funnel, helping identify user behavior patterns, engagement levels, and stages where potential customers exit the conversion process.

Key Highlights

- Displays overall touchpoints, Unique users, Average Touchpoints per User, Average Revenue per Conversion.
- Visualizes Customer Conversion funnel and Drop-off rates.
- Analyzes touchpoints by platform, medium and sources.
- Measures monthly touchpoints trend.
- Source level journey performance matrix.

Key Insights Derived

- Although campaigns successfully attract visitors, a significant proportion do not engage beyond the landing page. This indicates potential issues with landing page relevance, messaging, call-to-action placement, or audience targeting.
- Google is the strongest acquisition source, driving the highest customer engagement and generating the largest revenue contribution. This suggests that search-based traffic attracts highly valuable users.
- Paid Social generates the highest number of customer interactions, followed closely by Organic traffic.
- Customer engagement demonstrates seasonal variation, with stronger interaction levels during the later months of the year.

5. Overall Key Findings & Business Recommendations

Insight 1

First Touch Attribution generated the highest revenue (\$3.67M), followed by Linear (\$2.57M), while Last Touch contributed the least (\$2.10M). This indicates that awareness-stage marketing channels play the biggest role in initiating successful customer journeys.

Business Recommendation

Continue investing in top campaigns using multi-touch attribution rather than relying only on last-click attribution for budget allocation.

Insight 2

Paid Social consistently contributed the largest attributed revenue across all attribution models, outperforming Paid Search and Email.

Business Recommendation

Increase investment in Paid Social campaigns while continuously optimizing creatives and audience targeting to sustain performance.

Insight 3

Email Marketing achieved the highest ROAS (156.27) and the lowest CAC (\$2.47), making it the most cost-efficient acquisition channel.

Business Recommendation

Expand email automation, nurture campaigns, abandoned cart sequences, and personalized email journeys to maximize returns with minimal acquisition cost.

Insight 4

LinkedIn Ads recorded the highest Customer Acquisition Cost (\$21.26) while generating comparatively lower return than Email and TikTok campaigns.

Business Recommendation

Re-evaluate LinkedIn strategy and campaign objectives. Shift budget toward better-performing platforms unless LinkedIn supports high-value B2B objectives.

Insight 5

Campaign performance varies significantly even within the same platform. Campaigns such as TikTok_Gen_Z_Acquisition generated the highest revenue, while Email campaigns dominated ROAS rankings.

Business Recommendation

Allocate budget based on individual campaign performance rather than platform averages.

Insight 6

The customer journey experiences the largest drop-off immediately after the Page View stage. Only about half of visitors proceed to Click, with further decline toward final conversions.

Business Recommendation

Improve landing page quality and design, calls-to-action, page speed, and user experience to increase progression through the marketing funnel and reduce early-stage abandonment.

Insight 7

Google generated the highest number of customer touchpoints and also produced the highest total attributed revenue, demonstrating its importance in customer discovery.

Business Recommendation

Increase investment in Google Ads while maintaining a strong Search Engine Marketing (SEM) strategy. Continuously optimize keyword targeting and re-marketing campaigns to improve conversion performance.

Insight 8

Average customer interaction before conversion is approximately 1.27 touchpoints, while average revenue per conversion is around \$391. This suggests many conversions occur after very few recorded interactions.

Business Recommendation

Validate tracking implementation to ensure all touchpoints are captured accurately. If correct, focus on optimizing early customer interactions since they have the greatest influence on conversion outcomes.

Executive Summary

- **Top Revenue Attribution Model:** First Touch Attribution Model
 - **Best Performing Revenue Channel:** Paid Social
 - **Most Cost-Effective Platform:** Email Marketing
 - **Highest ROAS Campaigns:** Email Marketing campaigns
 - **Highest Revenue Campaign:** TikTok_Gen_Z_Acquisition
 - **Major Funnel Bottleneck:** Page View → Click transition
 - **Highest Traffic Source:** Google
 - **Primary Business Opportunity:** Reallocate marketing budget towards high ROAS channels while optimizing early-funnel engagement and reducing acquisition costs.
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6. Project Conclusion

Conclusion

The objective of this project was to develop an end-to-end **Multi-Touch Marketing Attribution and ROI Dashboard** that enables marketing teams to evaluate campaign performance beyond traditional last-click attribution. By integrating advertising spend, conversion, and customer journey data into a unified analytical model, the project provides a comprehensive view of marketing effectiveness across multiple channels and platforms.

The implementation of **First-Touch, Last-Touch, and Linear Attribution Models** demonstrated how attribution methodology significantly impacts revenue allocation and campaign evaluation. The analysis revealed that relying solely on a single attribution model may lead to inaccurate budget decisions, whereas multi-touch attribution provides a more balanced understanding of customer acquisition.

Overall, the project demonstrates how **Business Intelligence, SQL, Python, and Power BI** can be combined to convert fragmented marketing data into actionable insights, enabling organizations to optimize marketing investments, improve customer acquisition strategies, and maximize return on advertising spend.

Acknowledgement

I would like to express my sincere gratitude to **Infotact Solutions** for providing me with the opportunity to work on this project. This internship allowed me to strengthen my practical skills in Python, SQL, PostgreSQL, Power BI, and Data Analytics while gaining hands-on experience in solving real-world marketing attribution problems. I also extend my appreciation to everyone who contributed guidance and support throughout the successful completion of this project.

References

Software & Tools

- Microsoft Power BI Desktop
 - Power Query Editor
 - PgAdmin / PostgreSQL / SQL
 - Python (Pandas, Numpy, Matplotlib)
 - Jupyter Notebook
 - Microsoft Excel
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Concepts & Documentation

- Microsoft Power BI Documentation
 - SQL Window Functions and CTEs
 - Multi-Touch Attribution Models
 - Marketing Analytics & ROAS
 - Customer Acquisition Cost (CAC)
 - Conversion Funnel Analysis
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Dataset

Synthetic marketing datasets consisting of:

- Advertising Spend
- Customer Conversions
- User Journey / Touchpoints

used solely for educational and analytical purposes.
